DIRECT IMAGES OF RELATIVELY PERFECT COMPLEXES ALONG PROPER FLAT FINITELY PRESENTED MAPS OF SPACES ARE PERFECT WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12577-12585 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 12587-12612, the complete author proof (24 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. One of the two pillars of the chapter's relative-perfectivity toolkit (the other, descend-relatively-perfect, is the cited input); the proof's ordering — base change first because the ascent needs it — is preserved.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Derived Categories of Spaces, lemmas base-change-tensor and perfect-direct-image; Limits of Spaces, lemmas descend-finite-presentation, eventually-proper, descend-flat; the chapter's descend-relatively-perfect lemma; etale-locality of perfection on the target.

Source reference. Original lemma 0DKW: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12499–12503. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07SK: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12612–12612. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08K0: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12612–12612. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08K1: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12612–12612. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A1K: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12612–12612. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08IS: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12612–12612. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is flat, proper, and of finite presentation. Let $E \in D(\mathcal{O}_X)$ be Y-perfect. Then Rf_*E is a perfect object of $D(\mathcal{O}_Y)$ and its formation commutes with arbitrary base change.*

Proof. The statement on base change is Derived Categories of Spaces, Lemma 0A1K (with $\mathcal{G}^\bullet$ equal to $\mathcal{O}_X$ in degree 0). Thus it suffices to show that Rf_*E is a perfect object. We will reduce to the case where Y is Noetherian affine by a limit argument.

The question is étale local on Y, hence we may assume Y is affine. Say $Y = \mathrm{Spec}(R)$. We write $R = \mathrm{colim}\, R_i$ as a filtered colimit of Noetherian rings R_i. By Limits of Spaces, Lemma 07SK there exists an i and an algebraic space X_i of finite presentation over R_i whose base change to R is X. By Limits of Spaces, Lemmas 08K1 and 08K0 we may assume X_i is proper and flat over R_i. By Lemma 0DKW we may assume there exists a R_i-perfect object E_i of $D(\mathcal{O}_{X_i})$ whose pullback to X is E. Applying Derived Categories of Spaces, Lemma 08IS to $X_i \to \mathrm{Spec}(R_i)$ and E_i and using the base change property already shown we obtain the result. $\square$